Gaia-10

MEMBERSHIP

Stars in catalog	141
with Gaia GSP-Phot	65 (46%)
with radial velocity	26
Stream id (sID)	33

GEOMETRY · great-circle fit

Centroid RA, Dec	154.9, 15.5 deg
Centroid l, b	223.7, 52.3 deg
Pole RA, Dec	318.5, 73.9 deg
Length (φ1, 2–98%)	20.5 deg
Width σ(φ2), robust	0.35 deg

KINEMATICS

pmRA* med ± σ	-3.13 ± 0.82 mas/yr
pmDE med ± σ	-2.95 ± 0.33 mas/yr
μφ1* med ± σ	-2.91 ± 0.72 mas/yr
μφ2 med ± σ	-3.18 ± 0.42 mas/yr
Parallax (cat.) med	0.037 ± 0.166 mas
Parallax (wtd, ZP-corr.)	0.048 ± 0.013 mas
Distance	d = 20.9 ± 5.6 kpc
Physical length	7.48 kpc
RV median ± σ(MAD)	297.6 ± 4.4 km/s
RV gradient dV/dφ1	-0.75 km/s/deg

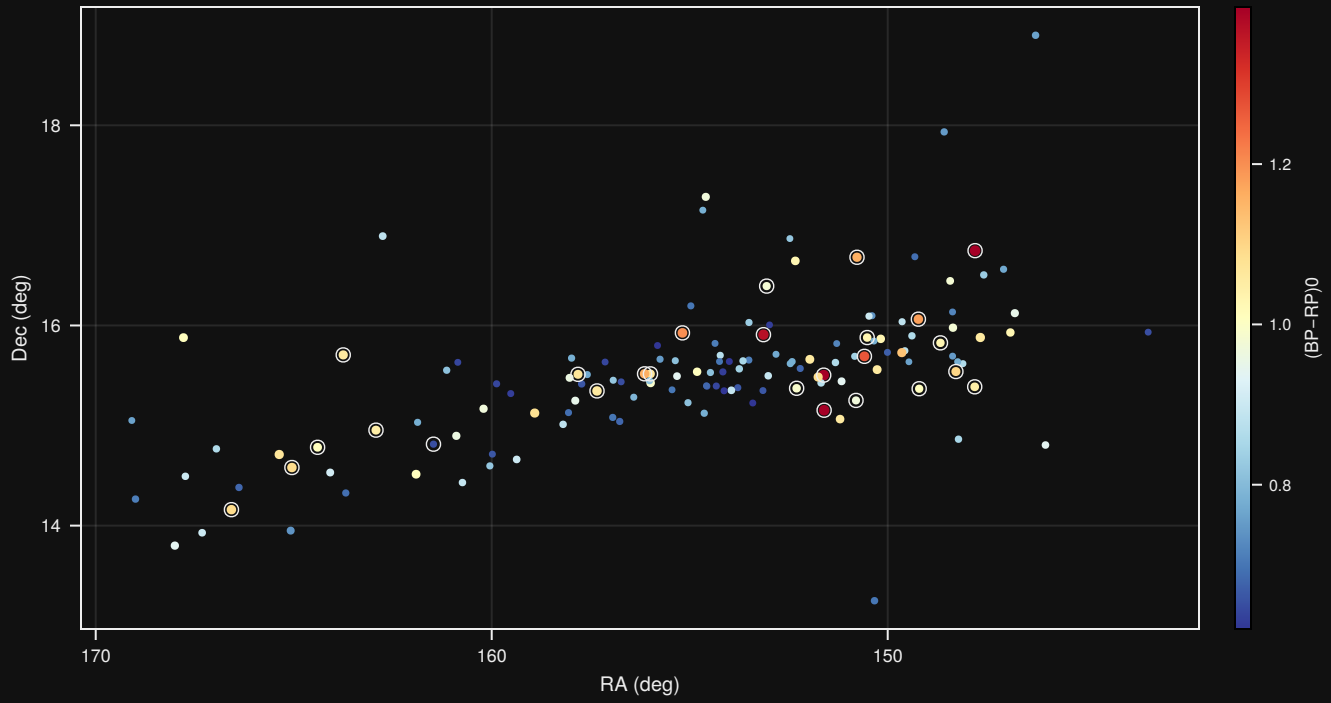
PHOTOMETRY · POPULATIONS

G0 range (median)	13.5–19.9 (18.9)
(BP−RP)0 median	0.85
GSP-Phot [M/H] (n=65)	-1.63 ± 0.61
GSP-Phot Teff median	5161 K
RUWE > 1.4	0.0 %
PM clusters (largest)	5 (38 stars)

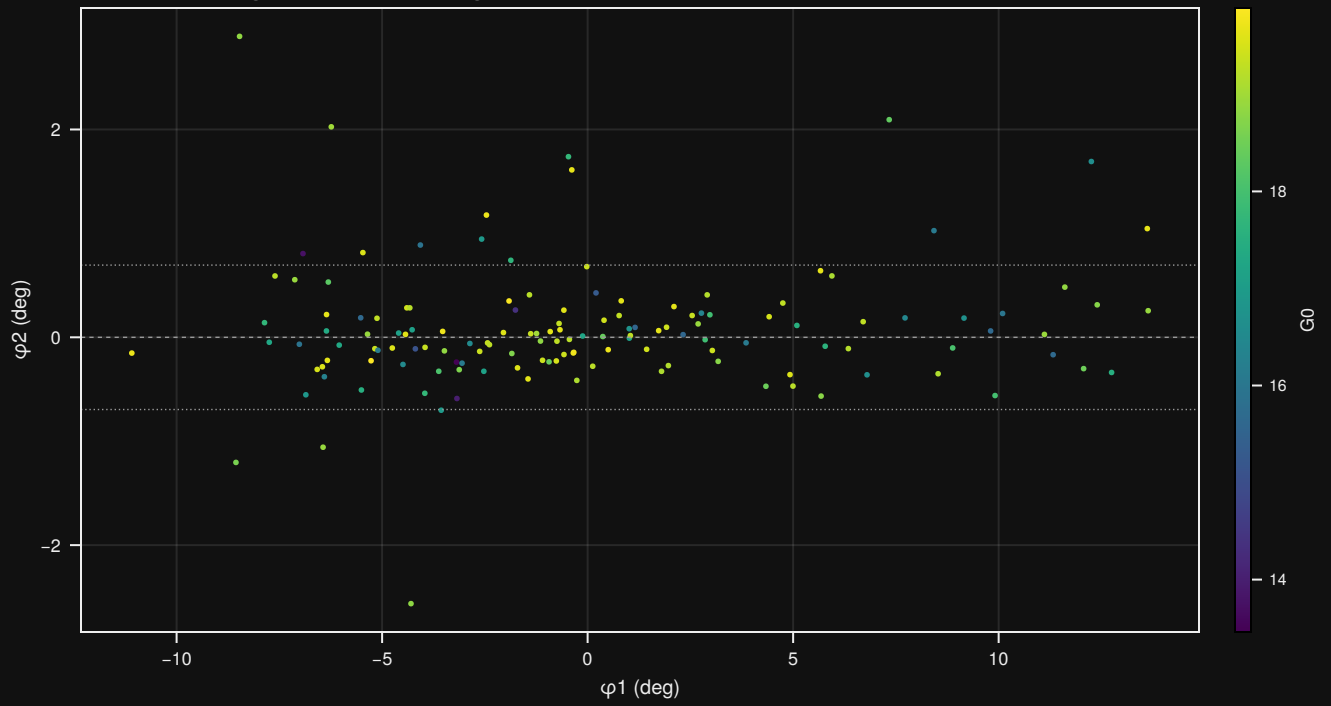
IBATA+2024 · TABLE 3

n catalog | paper	141 | 141
n_RV catalog | paper	26 | 26
<[Fe/H]> paper	-1.40
Stellar mass (paper)	7600 Msun
Pole offset vs paper	4.8 deg

Sky map · catalog members (n = 141) · size ∝ brightness · ring = has RV

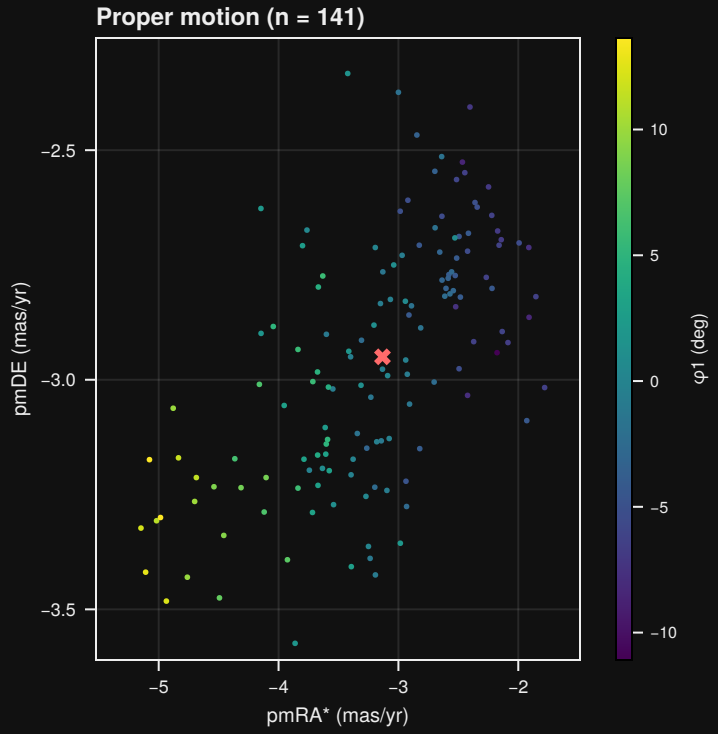


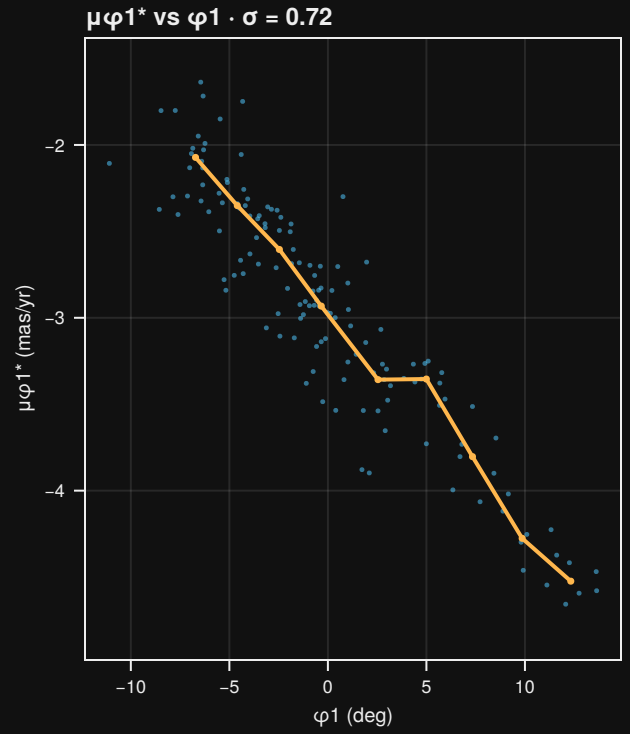
Stream frame (great-circle fit) · length 20.5° · width σ 0.35°

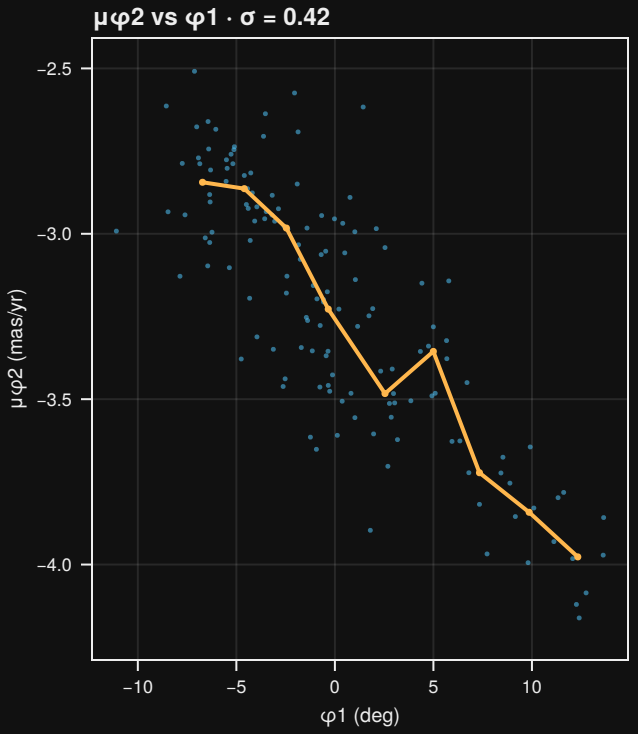


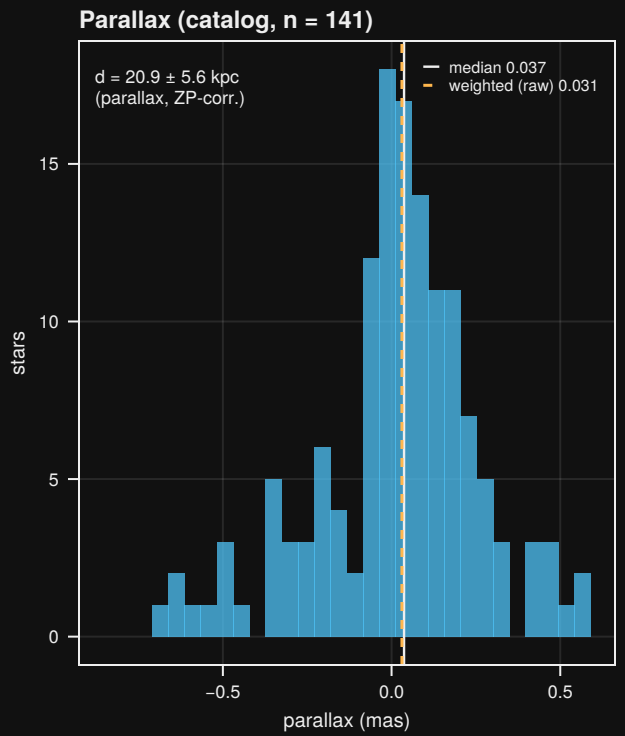
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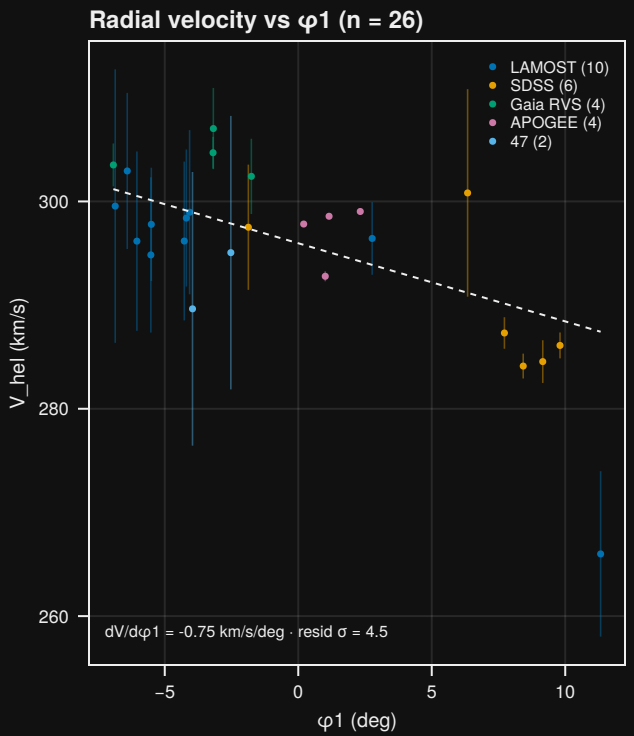
Kinematics · proper motion · parallax · radial velocity

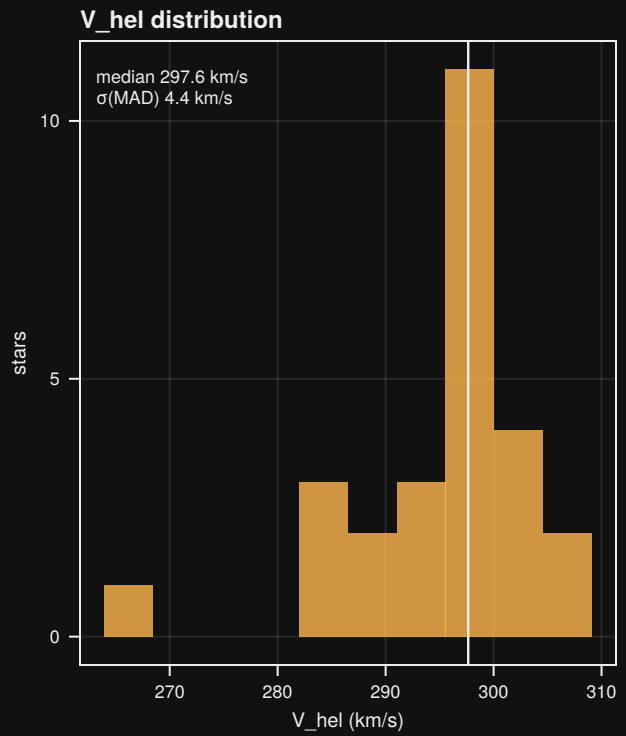






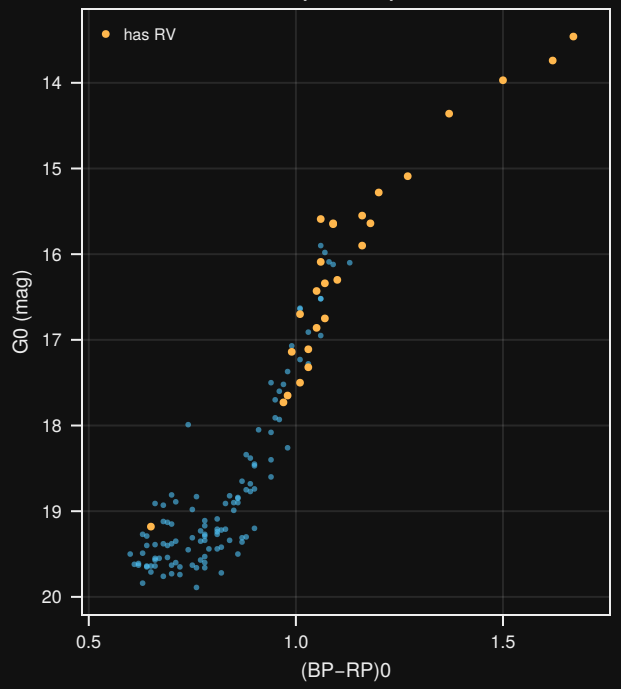




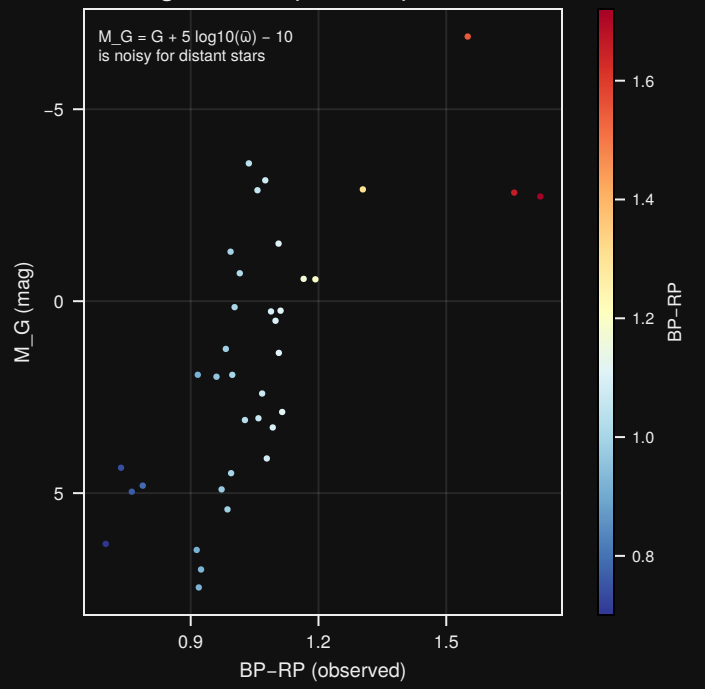


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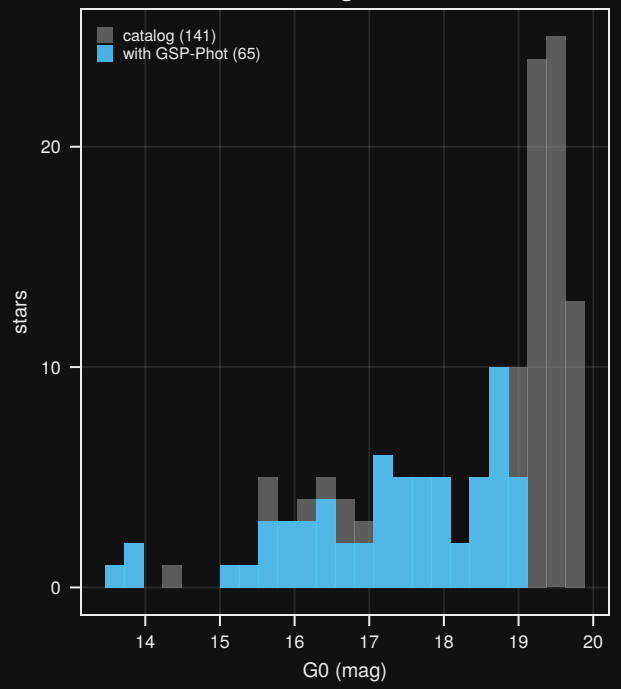
CMD · dereddened (n = 141)



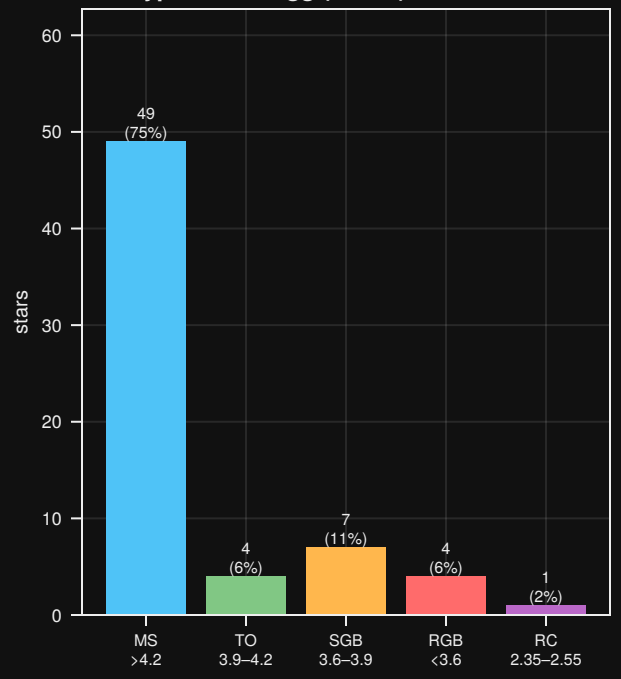
HR diagram · ϖ>0 (n = 37/65)



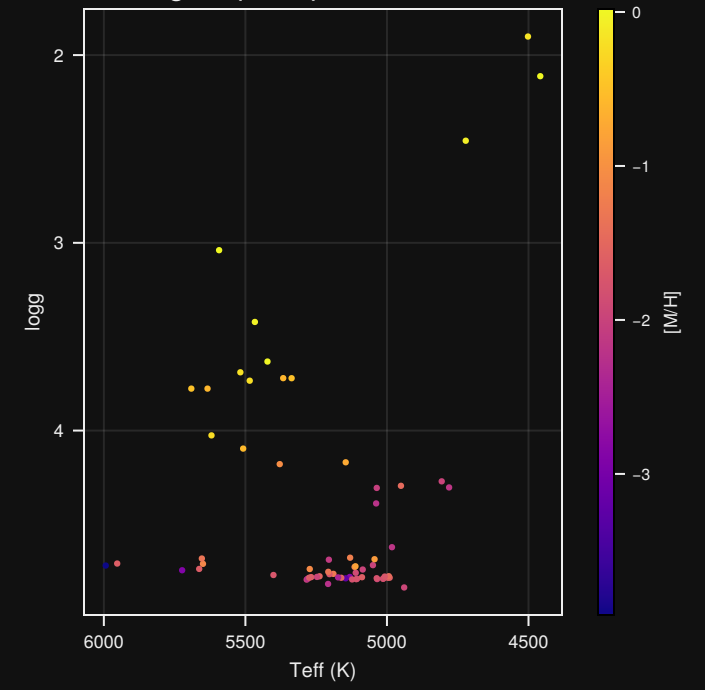
G0 & GSP-Phot coverage



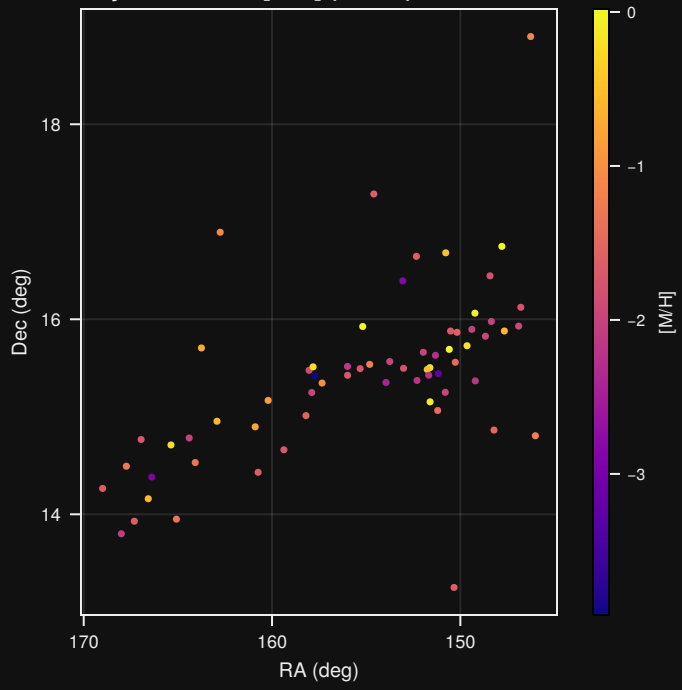
Star types from logg (n = 65)



Kiel diagram (n = 65)



Sky · GSP-Phot [M/H] (n = 65)

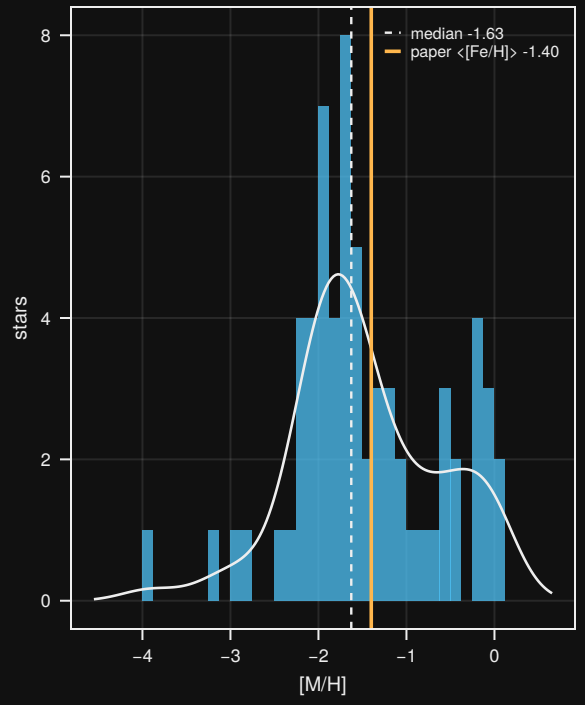


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

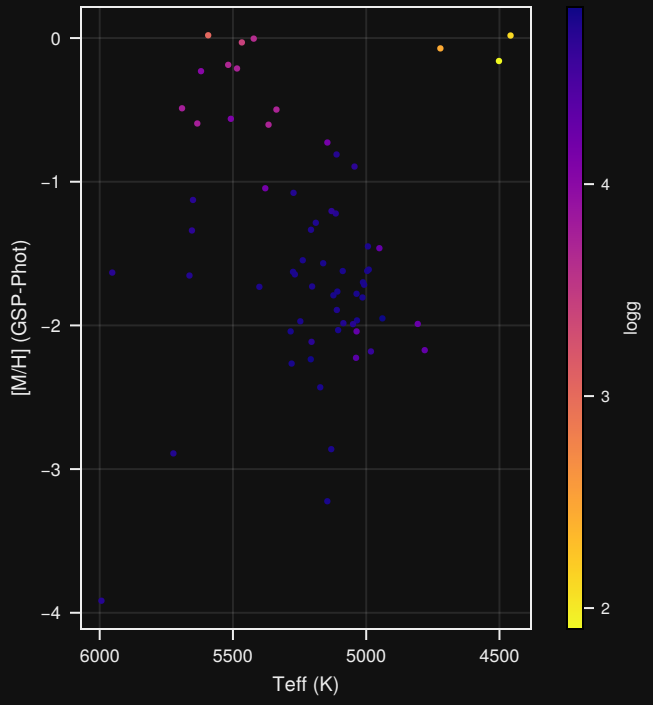
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Metallicity · age · astrometric quality

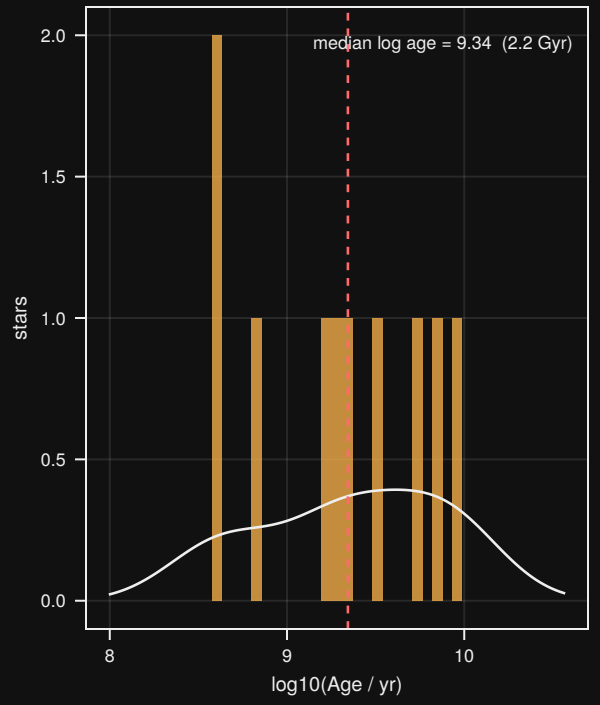
[M/H] GSP-Phot (n = 65)



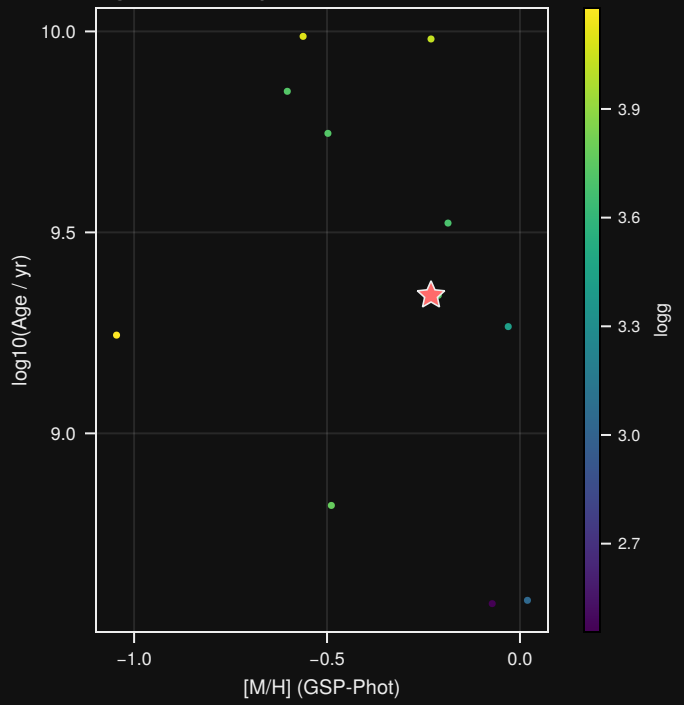
Teff vs [M/H] (n = 65)



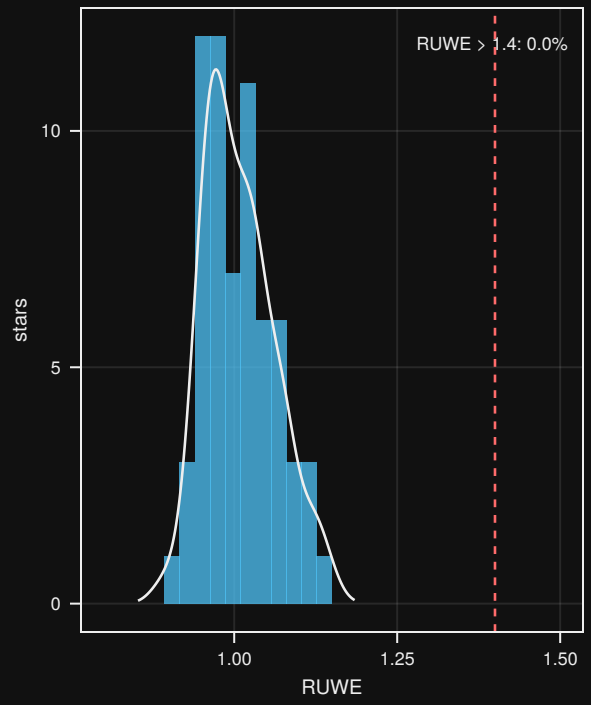
FLAME age (n = 11)



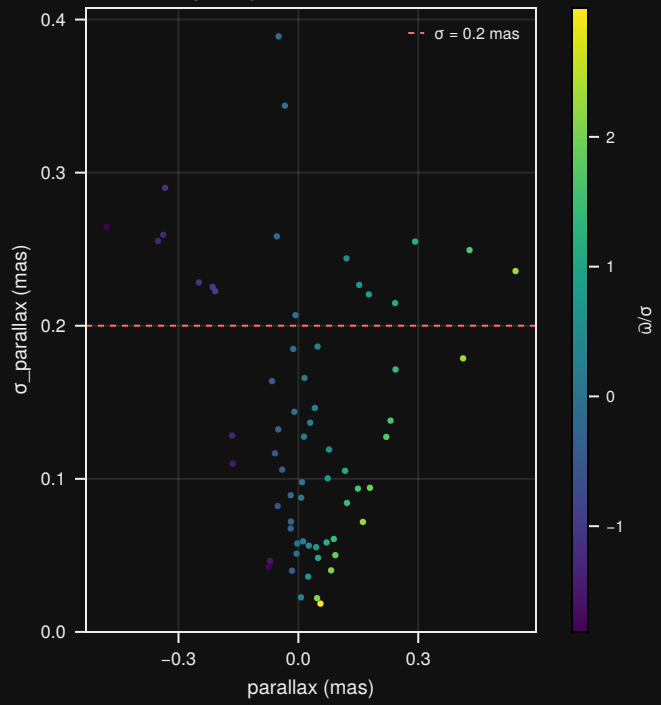
Age–metallicity (n = 11)



RUWE (n = 65)

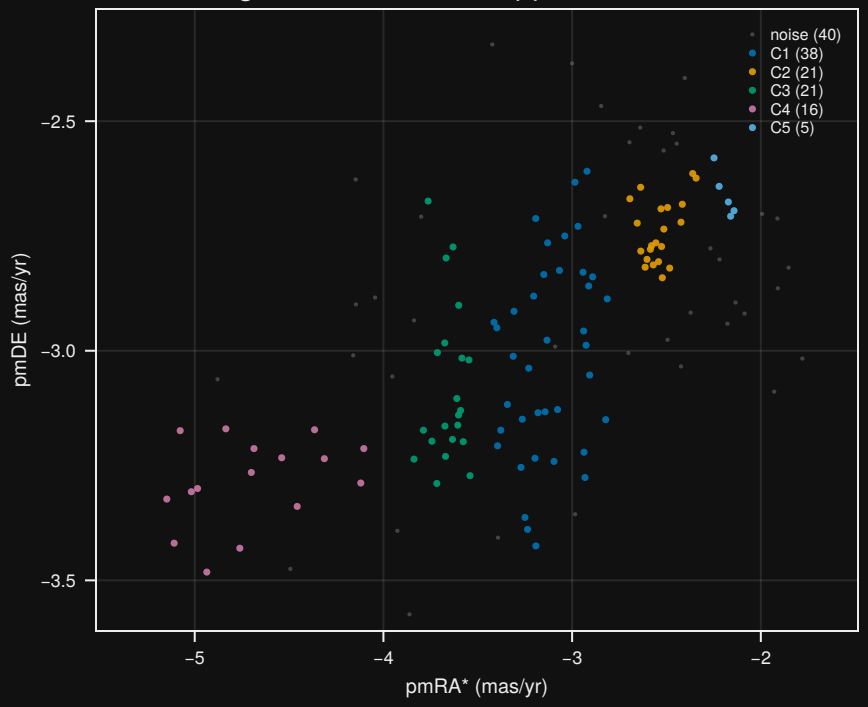


Parallax quality (n = 65)

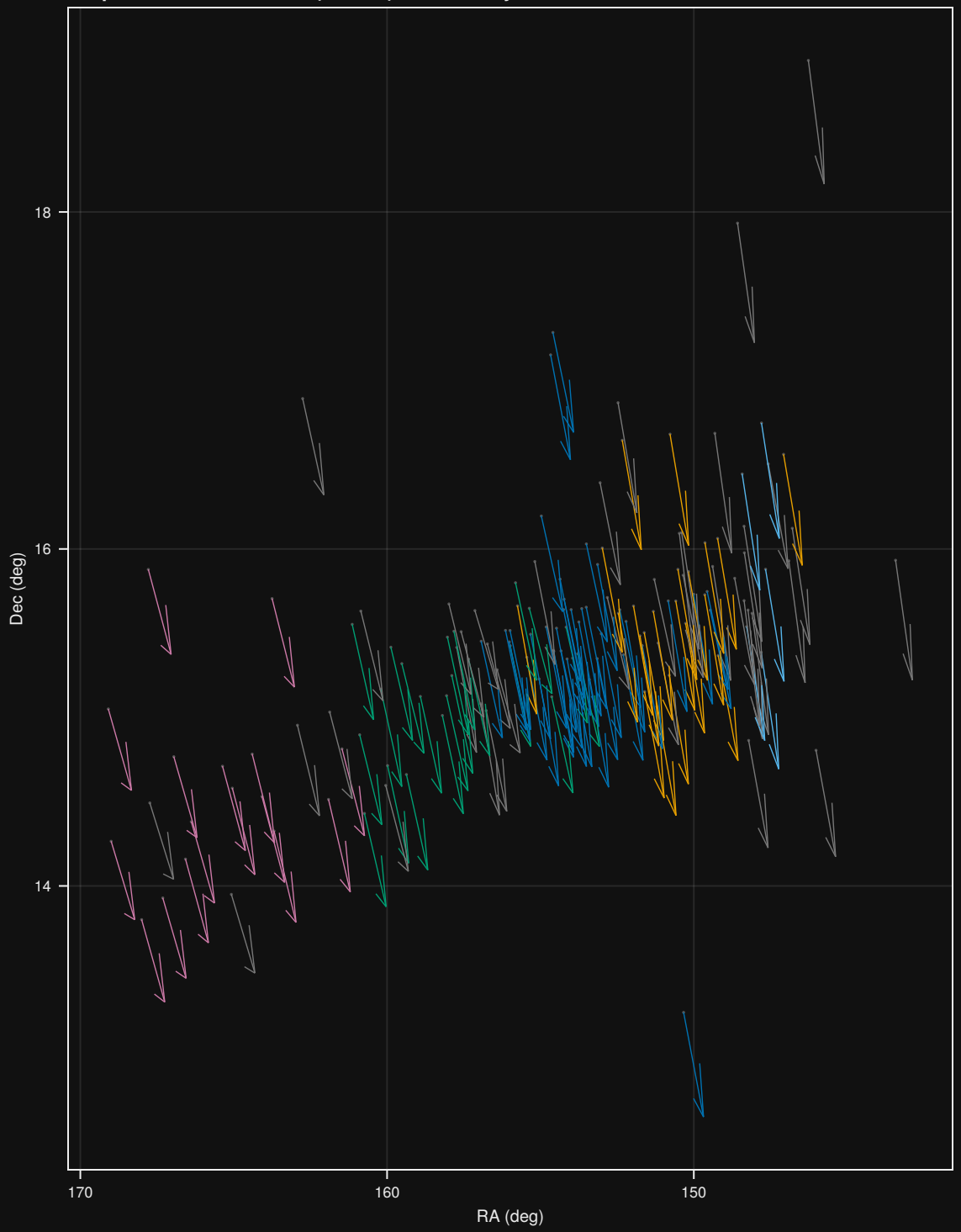


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PM clustering · HDBSCAN · 5 cluster(s) · noise 28%



Proper-motion directions (arrows) coloured by cluster



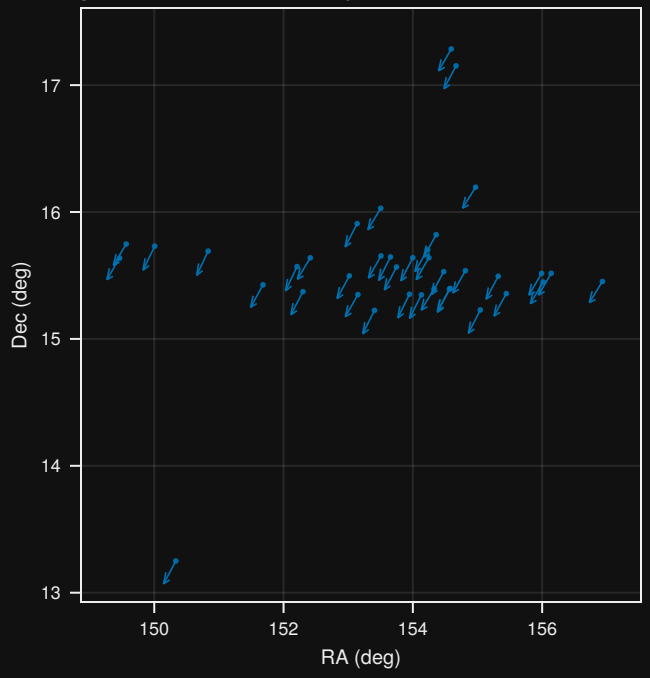
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	40	-2.70	-2.90	0.97	-1.66	4
C1	38	-3.14	-3.00	0.24	-1.88	4
C2	21	-2.54	-2.77	0.09	-1.46	8
C3	21	-3.63	-3.14	0.06	-1.60	2
C4	16	-4.73	-3.28	0.42	-1.65	5
C5	5	-2.17	-2.68	0.04	-0.59	3

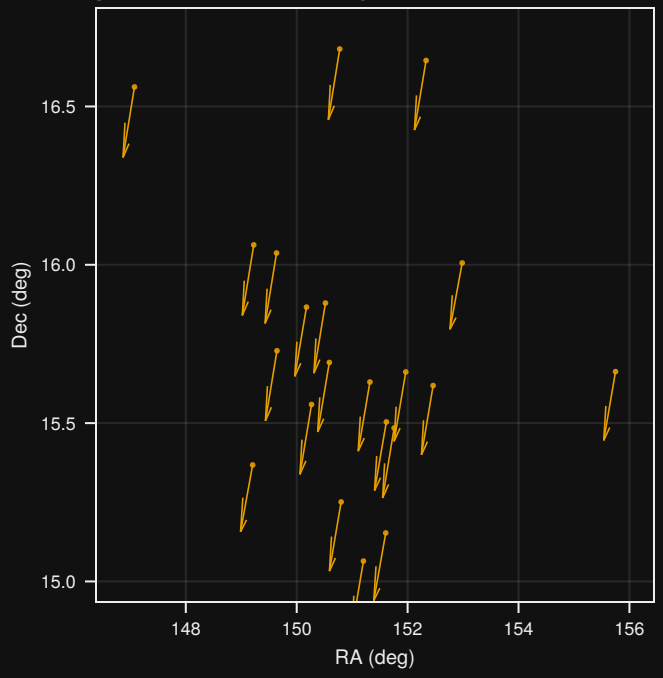
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Per-cluster proper-motion vector maps

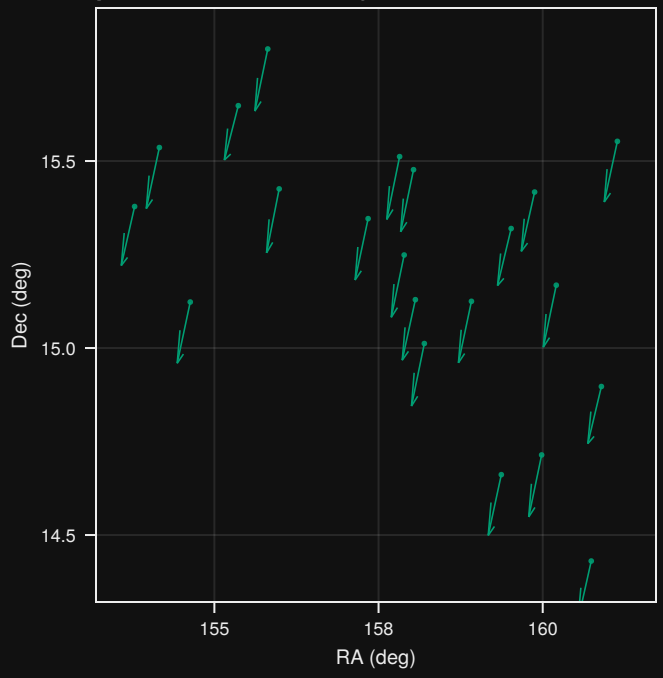
Cluster C1 · n = 38
pm = (-3.14, -3.00) mas/yr



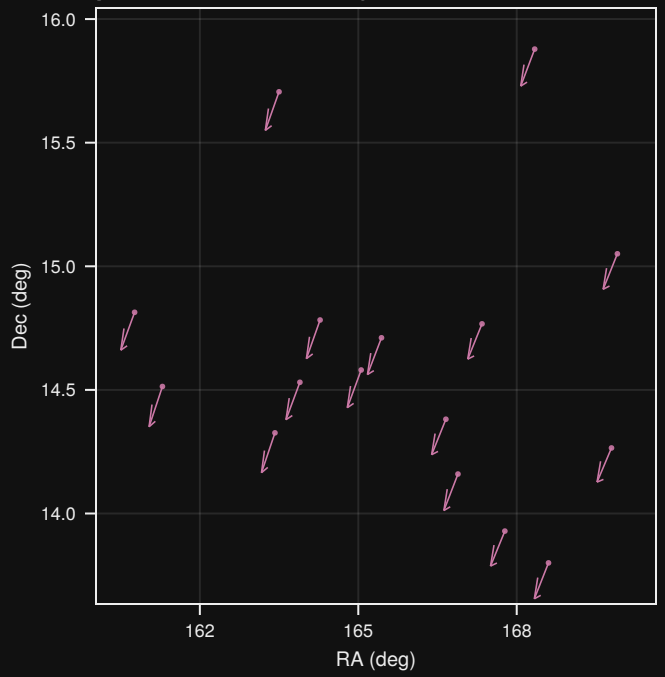
Cluster C2 · n = 21
pm = (-2.54, -2.77) mas/yr



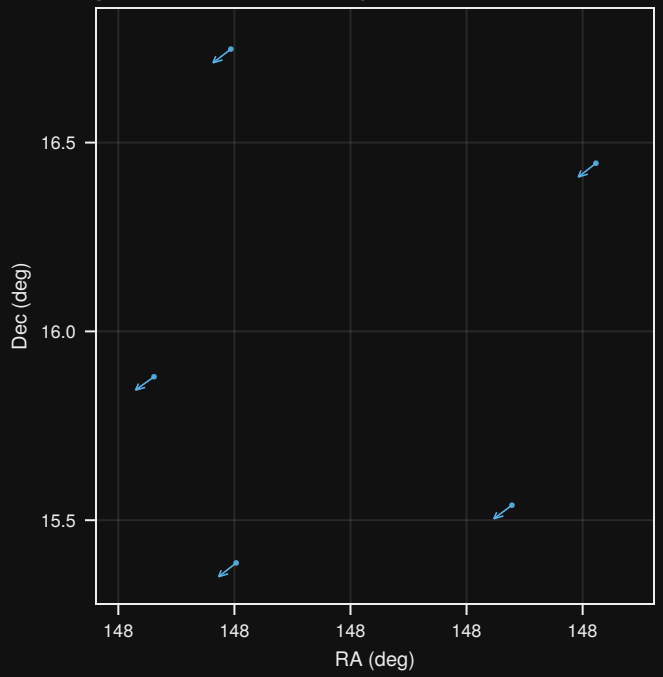
Cluster C3 · n = 21
pm = (-3.63, -3.14) mas/yr



Cluster C4 · n = 16
pm = (-4.73, -3.28) mas/yr



Cluster C5 · n = 5
pm = (-2.17, -2.68) mas/yr



Noise · n = 40

